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Esoteric Language Models

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Abstract

Diffusion-based language models offer a compelling alternative to autoregressive (AR) models by enabling parallel and controllable generation. Among this family of models, Masked Diffusion Models (MDMs) achieve the strongest performance but still underperform AR models in perplexity and lack key inference-time efficiency features—most notably, KV caching. In this work, we introduce Eso-LMs, a new family of models that fuses AR and MDM paradigms, enabling smooth interpolation between their perplexities while overcoming their respective limitations. Eso-LMs set a new state of the art on standard language modeling benchmarks. Crucially, we are the **first to introduce KV caching for MDMs** while preserving parallel generation, significantly improving inference efficiency. Combined with an optimized sampling schedule, our method achieves up to **65×** faster inference than standard MDMs and **4×** faster inference than prior semi-autoregressive approaches. We provide the code and model checkpoints on the project page:
https://s-sahoo.com/Eso-LMs

1 Introduction

We are at a pivotal moment in language modeling. A paradigm shift is underway—autoregressive (AR) language models, long considered the gold standard, are now being rivaled by diffusion language models for standard language generation. Recent works [26, 29] show that Masked Diffusion Models (MDMs) are closing the gap with AR models on small-scale language benchmarks, and even outperform them on tasks involving discrete structures, such as molecular generation [28, 11] and graph generation [14]. When scaled to larger sizes (e.g., 8B parameters), MDMs match models like LLaMA on challenging datasets in math, science, and tasks such as reverse poem completion [18].

These results make MDMs a compelling alternative to AR models. However, they suffer from two key limitations: (1) **Inference speed**: Despite supporting parallel generation, MDMs are significantly slower than AR models in practice, largely due to the lack of KV caching—a crucial optimization for real-time applications like chat systems. (2) **Generation quality**: MDMs still show a noticeable likelihood gap on more complex language modeling tasks [26].

Recently proposed BD3-LMs [1] address the speed issue by introducing a semi-autoregressive generation strategy. These models perform diffusion over fixed-length blocks of text sequentially. Because previously denoised blocks can be cached, BD3-LMs partially support KV caching and are faster than standard MDMs. However, we identify two key shortcomings in BD3-LMs: (1) **Mode collapse at low sampling steps**: When the number of denoising steps is reduced for faster inference, BD3-LMs exhibit severe degradation in sample quality—worse than both AR (at high Number of Function Evaluations (NFEs)) and diffusion models (at low NFEs) (Sec. 5.3). (2) **Incomplete**

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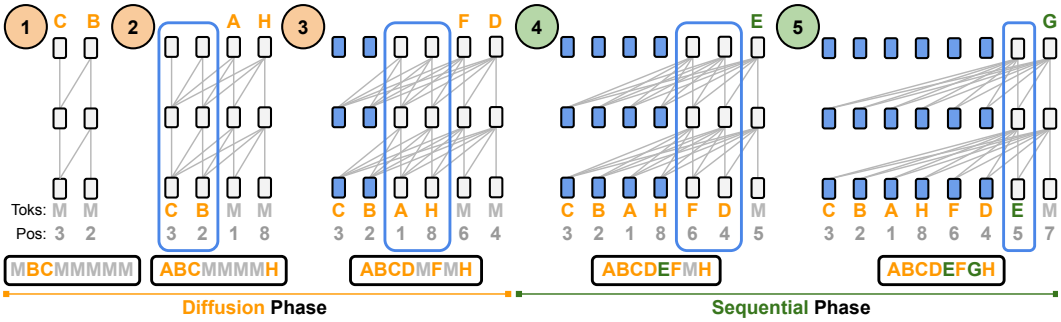


Figure 1: Efficient generation of an example sequence with our flagship model Eso-LM (B). During **Diffusion** Phase, Eso-LMs denoise one or more, potentially non-neighboring mask tokens (M) per step. During **Sequential** Phase, Eso-LMs denoise the remaining mask tokens one at a time from left to right. Eso-LM (B) allows for **KV caching in both phases** using just **a single unified KV cache**: **blue** bounding boxes enclose transformer cells that are building their KV cache; a cell becomes **blue** once its KV cache is built. The sequences below the transformers depict tokens in their natural order.

caching: While KV caching is possible across blocks, intra-block diffusion still lacks KV support, limiting overall speed gains.

To address these challenges, we propose a new language modeling paradigm that fuses autoregressive and masked diffusion approaches. Our model is trained with a hybrid loss—a combination of AR and MDM objectives—which allows it to interpolate smoothly between the two paradigms in terms of perplexity. This requires two key innovations: (1) A revised attention mechanism in the denoising transformer to support both AR and MDM styles of generation. (2) A new training and sampling procedure that enables KV caching within the diffusion phase, a feature previously unavailable in MDMs. Due to the unconventional nature of this hybrid design, we name our method **Eso**teric **L**anguage **M**odels (Eso-LMs).

Our method sets a new state of the art among diffusion models on language modeling benchmarks and enables fine-grained control over interpolation between AR and MDM perplexities. Unlike prior approaches, it supports KV caching during diffusion, yielding significantly faster inference. In summary, our contributions are as follows: (1) We propose a new framework for language modeling: one that fuses AR and MDM paradigms and outperforms the previous hybrid approach, BD3-LMs. (2) Ours is the **first approach to enable KV caching for MDMs** while preserving parallel generation, achieving up to $\mathbf{65}\times$ faster inference than standard MDMs and $\mathbf{3}-\mathbf{4}\times$ faster inference than KV-cached semi-autoregressive baselines (Sec. 5.2). (3) We show that BD3-LMs degrade at low sampling steps, while our method remains competitive with MDMs in the low NFE regime and with AR in the high NFE regime (Sec. 5.3). (4) Our results establish a new state-of-the-art perplexity for discrete diffusion, narrowing the gap to autoregressive models (Sec. 5.1).

2 Background

Notation We represent scalar discrete random variables that can take K values as 'one-hot' column vectors and define $\mathcal{V} \in \{\mathbf{x} \in \{0,1\}^K : \sum_{i=1}^K \mathbf{x}_i = 1\}$ as the set of all such vectors. Let $\mathbf{m} \in \mathcal{V}$ be a special mask vector such that $\mathbf{m}_K = 1$. Define $\text{Cat}(\cdot; \boldsymbol{\pi})$ as the categorical distribution over K classes with probabilities given by $\boldsymbol{\pi} \in \Delta^K$, where Δ^K denotes the K-simplex. Additionally, let $\langle \mathbf{a}, \mathbf{b} \rangle$ denote the dot product between vectors $\mathbf{a}$ and $\mathbf{b}$. We use parentheses $()$ to denote ordered sets (tuples) and curly brackets $\{\}$ to denote unordered sets. $|A|$ denotes the cardinality of the set A.

MDMs feature two salient orderings: the sequence order and the denoising order. We use a permutation σ to describe the relationship between these orderings. Let $\mathcal{P}_L$ denote the set of all permutations of $[L] = \{1, \dots, L\}$. $\sigma \in \mathcal{P}_L$ is an ordered set (tuple) and also serves as a bijective function: $\sigma(l)$ is the item at the lth position in σ, and $\sigma^{-1}(i)$ is the position of the item i in σ. For example, $(2, 4, 1, 3)$ denotes an ordering of $\{1, 2, 3, 4\}$, with $\sigma(2) = 4$ mapping the position 2 to its corresponding item 4, and $\sigma^{-1}(4) = 2$ mapping the item 4 to its corresponding position 2.

Let $\mathbf{x} \in \mathcal{V}^L$ denote a sequence of length L with no mask tokens, and let $\mathbf{x}^\ell$ denote the ℓth entry in $\mathbf{x}$. We use the term 'token index' to refer to the position of a token in the original ordering, e.g., the token index for $\mathbf{x}^l$ is l. Let $(\mathbf{z}_t)_{t\in[0,1]} \in \mathcal{V}^L$ denote a sequence of length L that may contain mask tokens. Let $\mathcal{M}(\mathbf{z}_t) = \{\ell \mid \mathbf{z}_t^\ell = \mathbf{m}\}$ denote token indices of mask tokens in $\mathbf{z}_t$ and $\mathcal{C}(\mathbf{z}_t) = \{\ell \mid \mathbf{z}_t^\ell \neq \mathbf{m}\}$ denote token indices of clean tokens in $\mathbf{z}_t$.

Let $\oplus : \mathcal{V}^m \times \mathcal{V}^n \to \mathcal{V}^{m+n}$ denote a concatenation operator on two sequences $\mathbf{x} = (\mathbf{x}^1, \mathbf{x}^2, \ldots, \mathbf{x}^m)$ and $\mathbf{z} = (\mathbf{z}^1, \mathbf{z}^2, \ldots, \mathbf{z}^n)$ of length m and n. When the concatenated sequence $\mathbf{x} \oplus \mathbf{z}$ is fed into the transformer, $\mathbf{x}$ and $\mathbf{z}$ carry the same positional embeddings as they would if they were fed into a transformer independently. Let $\odot : \mathcal{V}^m \times \mathcal{V}^n \to \mathcal{V}^m$ denote a substitution operator; for any $\mathbf{z} \in \mathcal{V}^m$ and $\mathbf{x} \in \mathcal{V}^n$ with $m > n$, the output $\mathbf{y} = \mathbf{z} \odot \mathbf{x}$ is given by: $\mathbf{y}^{1:n} = \mathbf{x}$ and $\mathbf{y}^{n+1:m} = \mathbf{z}^{n+1:m}$.

2.1 Autoregressive Models

Given a sequence $\mathbf{x} \sim q_{\text{data}}$ in $\mathcal{V}^L$, autoregressive (AR) models define the following factorization of the joint distribution: $\log p_\theta(\mathbf{x}) = \sum_{\ell=1}^{L} \log p_\theta(\mathbf{x}^\ell \mid \mathbf{x}^{<\ell})$, where the model p_θ is usually parameterized by a causal transformer [34] model. Sampling takes L steps but is efficient due to KV caching. AR models achieve the best likelihood and generation quality.

2.2 Masked Diffusion Models

Diffusion models learn to reverse a forward corruption process q, which transforms clean data $\mathbf{x} \sim q_{\text{data}}$ in $\mathcal{V}^L$ into a sequence of latent variables $\mathbf{z}_t$ for $t \in [0, 1]$, each representing an increasingly noisy version of $\mathbf{x}$ [8, 31, 32]. In MDMs [26, 30, 20], the forward masking process q factors independently across the sequence $\mathbf{x}$, i.e., $q_t(.|\mathbf{x}) = \Pi_\ell q_t(.|\mathbf{x}^\ell)$ and each token $\mathbf{x}^l$ is progressively interpolated with a fixed target distribution $\text{Cat}(.; \mathbf{m})$. The marginal of $\mathbf{z}_t^\ell \sim q_t(.|\mathbf{x}^\ell)$ at time t is given by:

$$q_t(.|\mathbf{x}^\ell) = \text{Cat}(.; \alpha_t \mathbf{x}^\ell + (1 - \alpha_t)\mathbf{m}), \tag{1}$$

where $\alpha_t \in [0, 1]$ is a strictly decreasing function in t with $\alpha_0 \approx 1$ and $\alpha_1 \approx 0$. Sahoo et al. [26] show that the reverse posterior $q_{s|t}(.|\mathbf{z}_t^\ell, \mathbf{x}^\ell)$ over $\mathbf{z}_s^\ell$ for $s < t$ is given by

$$q_{s|t}(.|\mathbf{z}_t^\ell, \mathbf{x}^\ell) = \begin{cases} \text{Cat}(.; \mathbf{z}_t^\ell) & \mathbf{z}_t^\ell \neq \mathbf{m}, \\ \text{Cat}\left(.; \frac{(1-\alpha_s)\mathbf{m} + (\alpha_s - \alpha_t)\mathbf{x}^\ell}{1-\alpha_t}\right) & \mathbf{z}_t^\ell = \mathbf{m}. \end{cases} \tag{2}$$

Given a denoising model $\mathbf{x}_\theta : \mathcal{V}^L \to (\Delta^K)^L$, the reverse unmasking process $p_\theta(.|\mathbf{z}_t)_{s|t}$ over the sequence $\mathbf{z}_s$ is parameterized by

$$p_\theta(.|\mathbf{z}_t)_{s|t} = \prod_{\ell}^{L} p_\theta^\ell(.|\mathbf{z}_t)_{s|t} = \prod_{\ell}^{L} q_{s|t}^\ell(.|\mathbf{z}_t^\ell, \mathbf{x}^\ell = \mathbf{x}_\theta^\ell(\mathbf{z}_t)). \tag{3}$$

Sahoo et al. [26], Shi et al. [30], Ou et al. [20] show that Negative Evidence Lower Bound (NELBO) for this method is

$$\mathcal{L}_{\text{MDM}}(\mathbf{x}) = \mathbb{E}_{q, t\sim[0,1]} \frac{\alpha_t'}{1 - \alpha_t} \sum_{\ell \in \mathcal{M}(\mathbf{z}_t)} \log \langle \mathbf{x}_\theta^\ell(\mathbf{z}_t), \mathbf{x}^\ell \rangle, \tag{4}$$

which is a weighted average of masked language modeling losses [5] computed only on the masked tokens $\mathcal{M}(\mathbf{z}_t)$.

To generate a sequence of length L, the reverse diffusion process starts from $\mathbf{z}_{t=1}$, where $\mathbf{z}_{t=1}^\ell = \mathbf{m}$ for $\ell = 1, \ldots, L$. It proceeds for T steps, with each $\mathbf{z}_s^\ell$ independently sampled from $p_\theta(.|\mathbf{z}_t)_{s|t}$ as defined in (3); once a position is unmasked, it remains fixed. Since multiple tokens can be denoised in parallel, the total number of network function evaluations (NFEs) can be less than L, enabling faster generation. However, each NFE is expensive due to the bidirectional transformer in $\mathbf{x}_\theta(\mathbf{z}_t)$.

2.3 Block Denoising Diffusion Discrete Language Models

Block Denoising Diffusion Discrete Language Models (BD3-LMs) [1] autoregressively model blocks of tokens and perform masked diffusion modeling (Sec. 2.2) within each block. By changing the